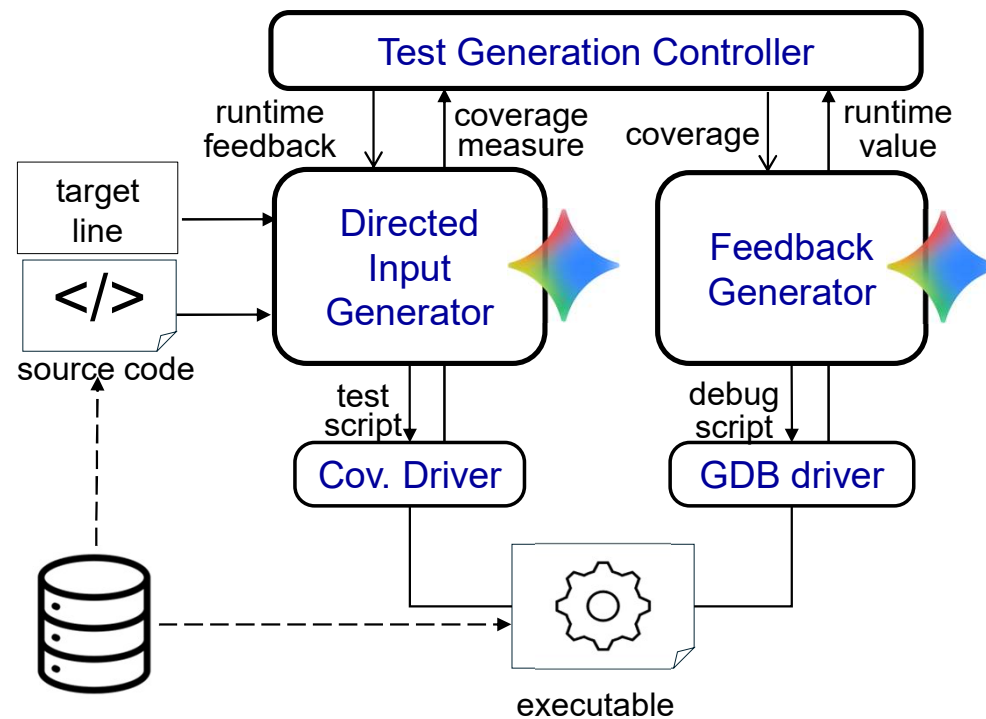


Runtime Value Feedback-guided LLM-based Directed Test Input Generation

Narin Han · Shin Hong (충북대학교)

Runtime Feedback-guided Refinement

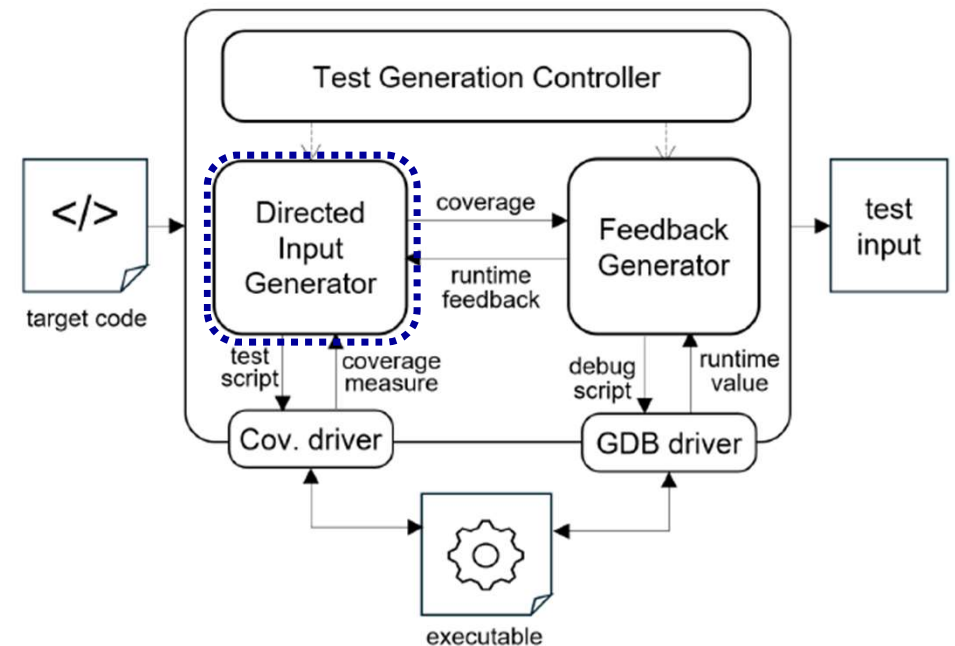
- Generate a test input, and iteratively refine the input toward target
- Use LLM to diagnose and fix the problem in target-missing test input



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

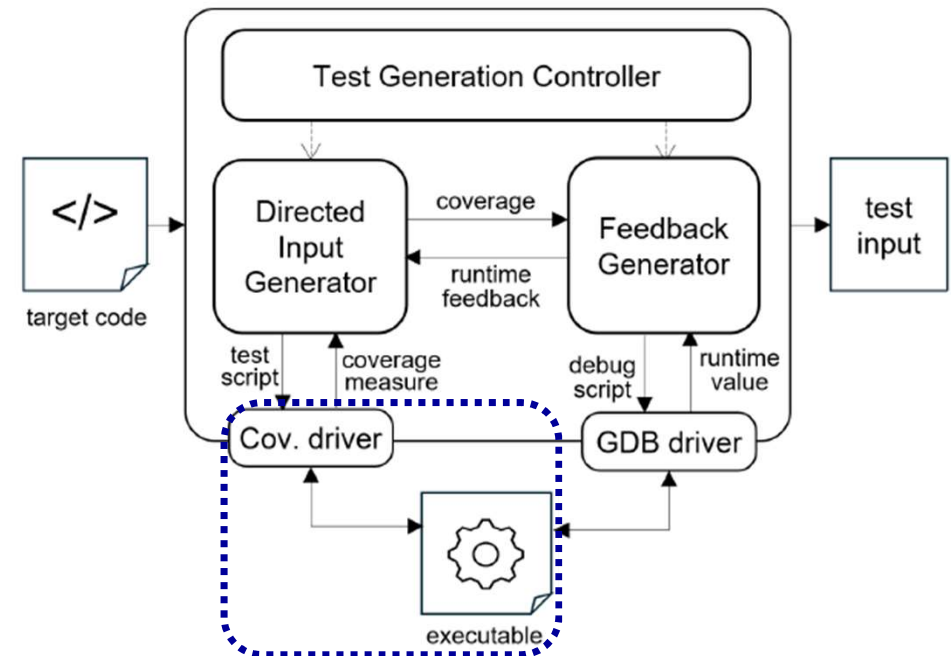
Turn 1. Directed Input Generation



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

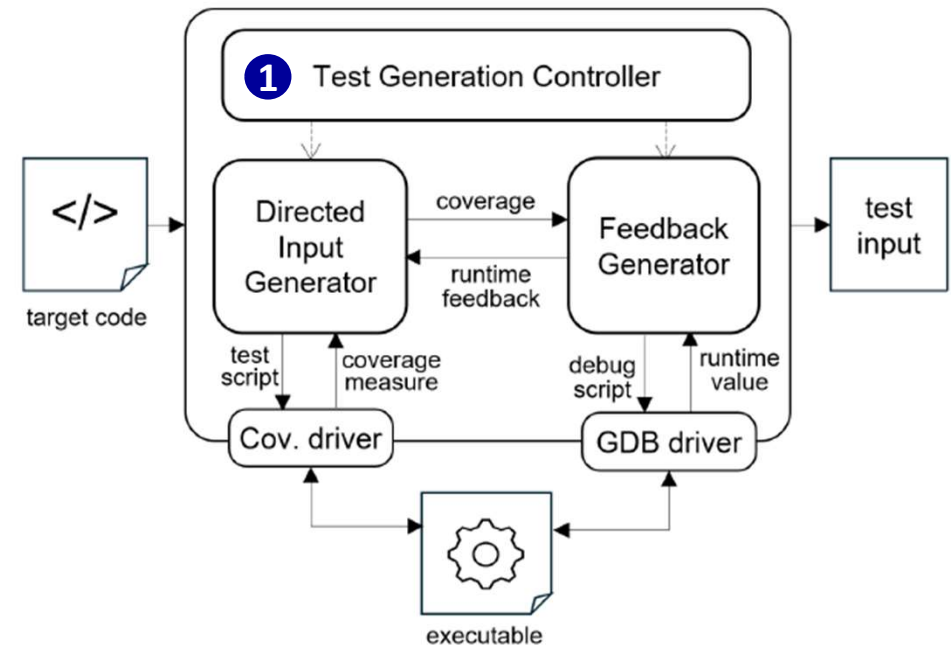
Turn 1. Directed Input Generation ➔ Target missed



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

Turn 1. Directed Input Generation
➔ Target missed ➔ Define Seed

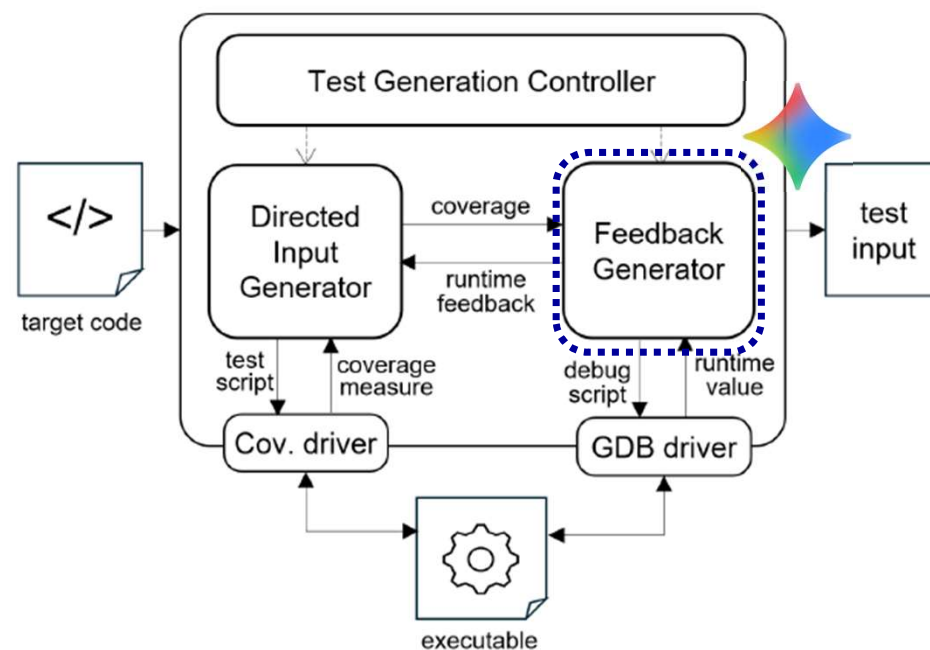


Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634 if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637 Unicode cCurrent = getCombiningChar(u);
638 Unicode cPrev = getCombiningChar(text[len - 1]);
...
642 if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

Turn 2. Feedback Generator

➔ Request runtime value queries

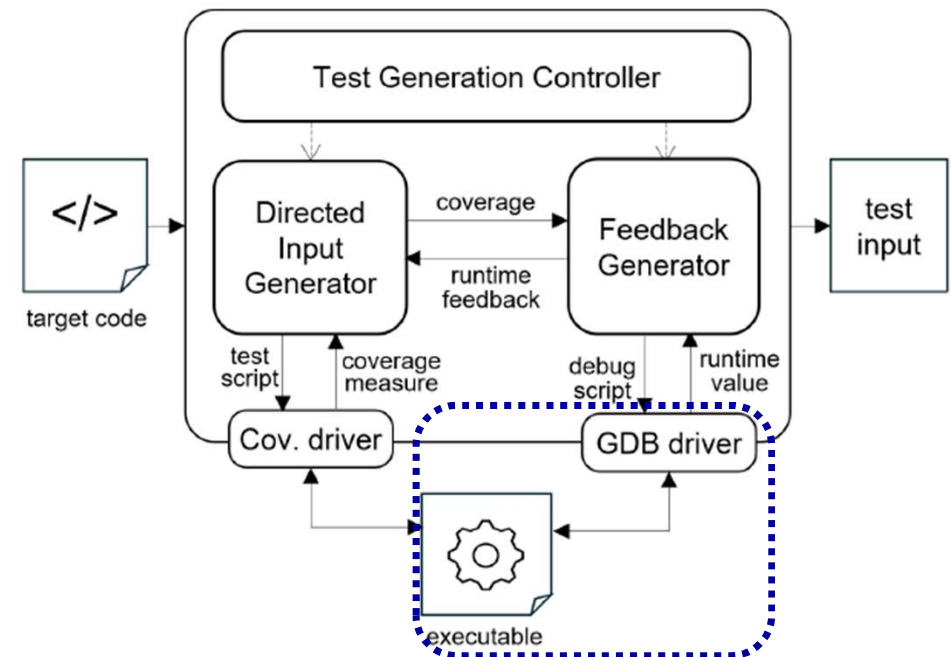


Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634 if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return
...
637 Unicode cPrev = getCombiningChar(u);
638 Unicode cPrev = getCombiningChar(text[len - 1]);
...
642 if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

len: 0

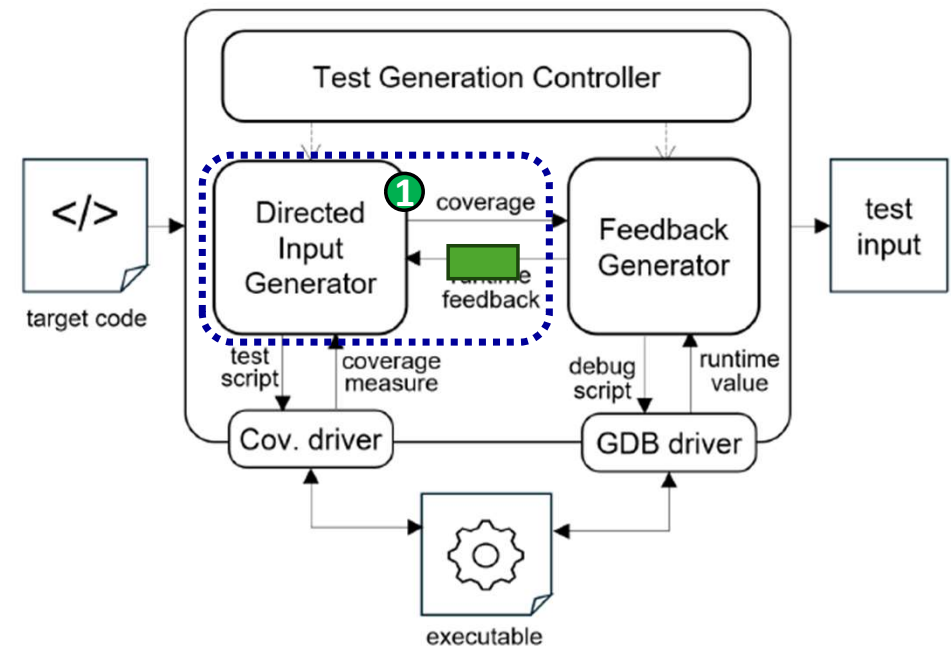
Turn 2. Feedback Generator ➔ Extract runtime values



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634 if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return
...
637 Unicode cPrev = getCombiningChar(u);
638 Unicode cPrev = getCombiningChar(text[len - 1]);
...
642 if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

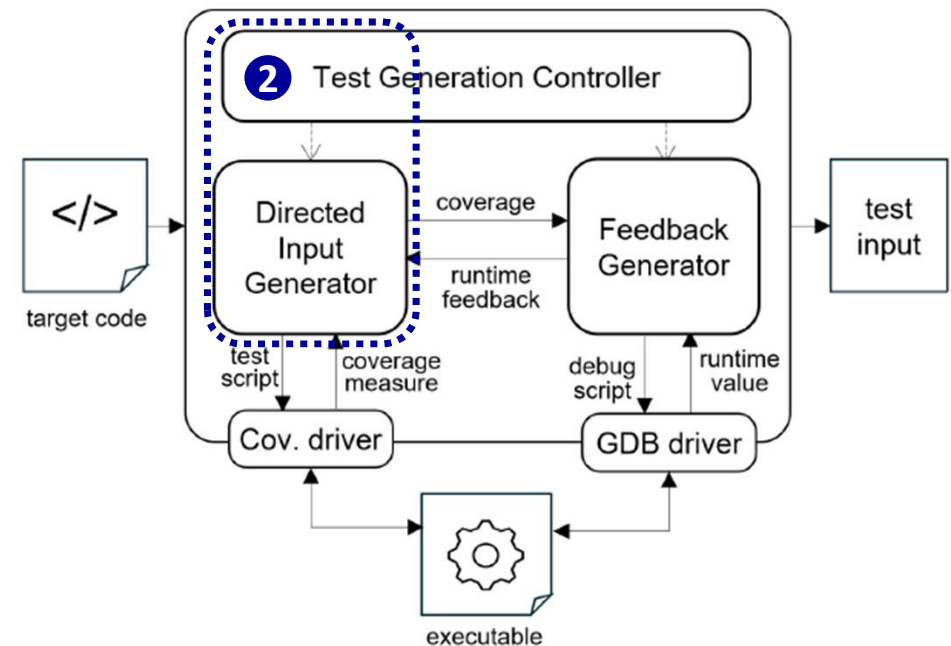
Turn 3. Directed Input Generation with Seed and Feedback



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634 if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return
...
637 Unicode cPrev = getCombiningChar(u);
638 Unicode cPrev = getCombiningChar(text[len - 1]);
...
642 if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

Turn 3. Directed Input Generation with Seed and Feedback → **Update Seed**

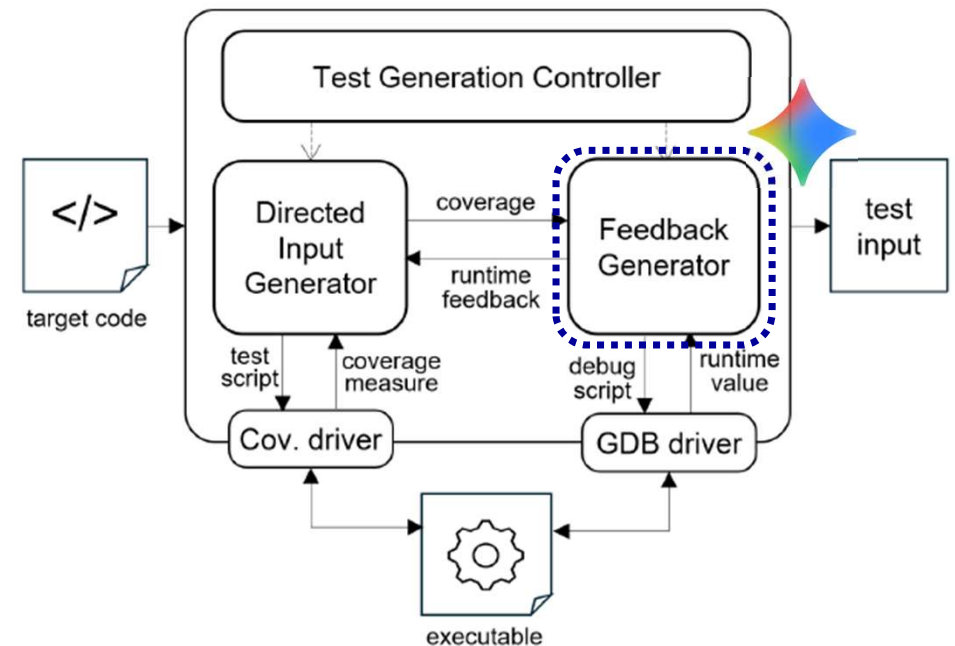


Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

Turn 4. Feedback Generation

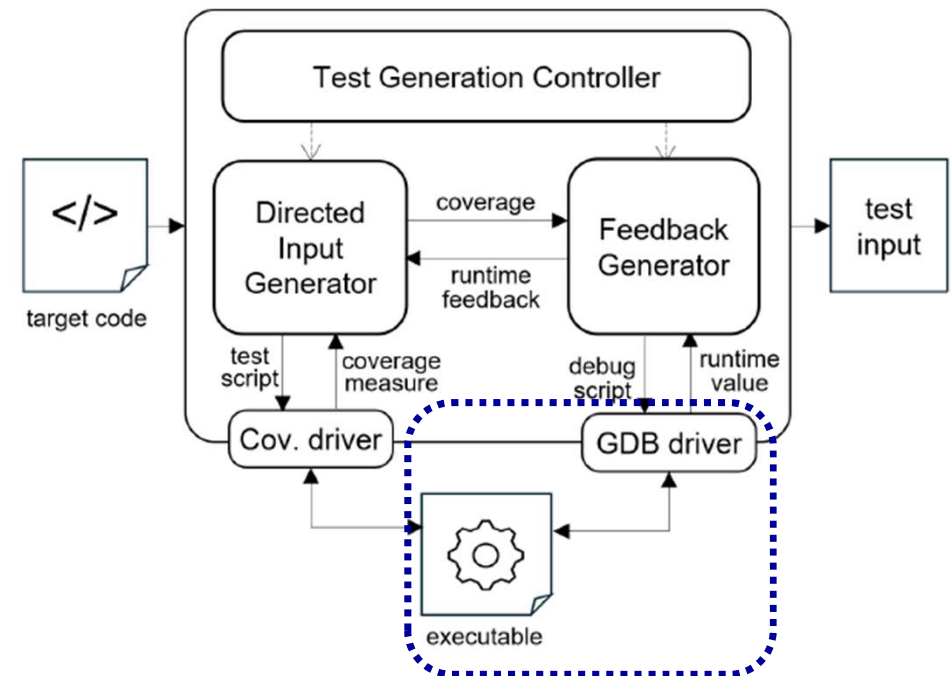
➔ Request runtime value queries



Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && unicodeTypeAlphaNum(text[len - 1])){
        cPrev: 0
...
676   if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
678   maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

Turn 4. Feedback Generation ➔ Extract runtime values

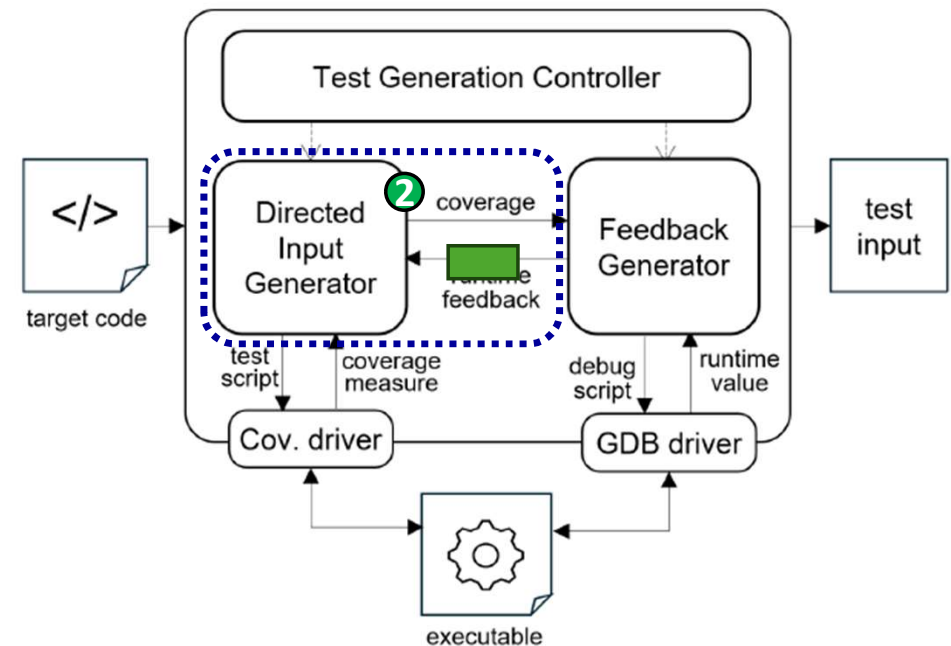


Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && !AlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

cPrev: 0

Turn 4. Directed Input Generation with Seed and Feedback

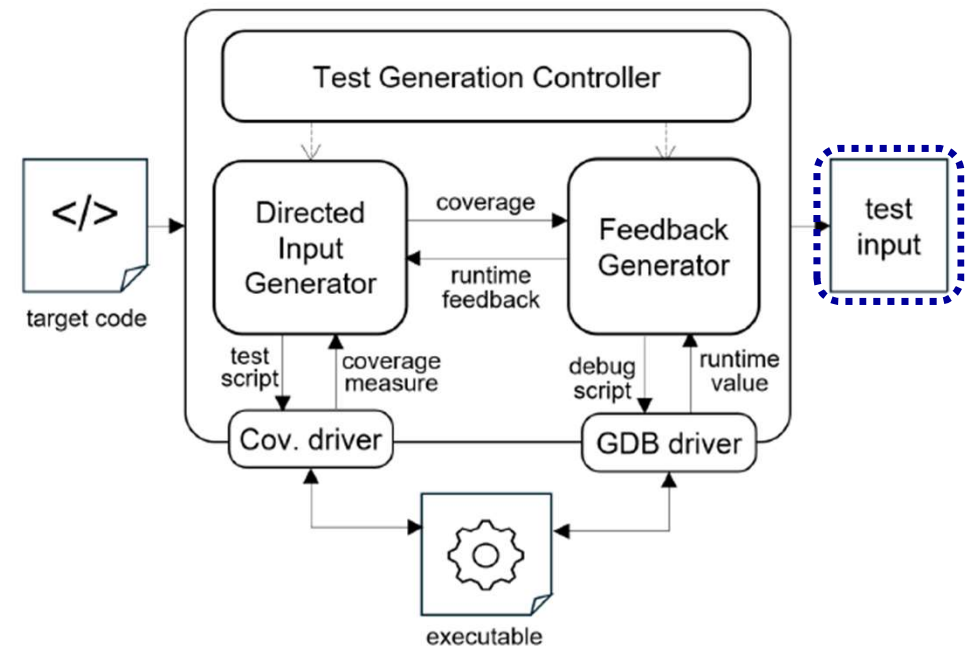


Example: addCombining() in Poppler

```
632 addCombining (FontInfo *fA, double fsA, Unicode u,...)
634   if (len == 0 || wMode != 0 || fA->getWMode() != 0)
635     return;
...
637   Unicode cCurrent = getCombiningChar(u);
638   Unicode cPrev = getCombiningChar(text[len - 1]);
...
642   if(cCurrent != 0 && !AlphaNum(text[len - 1])){
...
676     if(cPrev != 0 && unicodeTypeAlphaNum(u)) {
...
678     maxD=(fA->getAsc()-fA->getDest())*fsA; /**TARGET**/
```

cPrev: 0

Turn 4. Directed Input Generation with Seed and Feedback



Case Studies

- Implemented the framework with LLVM 9, GDB 8.1 and GPT-5.4
 - additionally provide the function code in a call stack
- Construct four case study subjects

Subjects	Success	Tests	Turns	Feedback
Poppler:addCombining:644	Yes	11	2	5
Poppler:addCombining:678	Yes	13	2	5
Libsndfile:ogg_opus_read_refill:982	Yes	7	2	5
Libsndfile:ogg_opus_read_refill:1062	No	15	7	15

Future Works & Discussions

- Exploring alternative feedback-generation strategies
- Reusing previously generated seeds and feedback
- Adaptive search control
- Comprehensive empirical evaluation
- What are appropriate to target?

**Runtime Feedback-guided
LLM-based Directed
Test Input Generation**



Narin Han

**Knowledge-driven Discovery of
Issue-Commit Links in Fuzzing
History**



Byul Yoon



Yuntae Kim

**Evaluating RLM-powered
SWE-Agent in
Refactoring Tasks**



Joonhuei Kim